iQ3R Mathematics — Geometry[1]
Lesson 7 Module 1 [2]

June 11, 2022

This 7th lesson presents a combination of materials from the New York Curriculum which are introduced in their lesson 6 and lesson 8. This is the first official departure from topic A and beginning of topic B materials. Students will review formerly learned geometry facts and practice citing the geometric justifications in anticipation of unknown angle proofs.

[1] Typically in the United States grade 9 students take either Algebra 1 or Geometry depending on the school and the performance of the student in 8th grade. Generally, Geometry is taken by students who are on a more accelerated track.

[2] Adapted for individualized instruction from the EngageNY curriculum for New York State public and private instructors.

Student View: Slide 1

Overview

IN THIS LESSON you will review some of the material you previously studied in earlier grades on angles. Specifically, the focus will be on solving for unknown angles — lines at a point and triangles. In each of the cases we will review how to find unknown angles.

1. When lines intersect at a point
2. When n angles lie on a line
3. When rays share a common vertex at a point
4. Angles in triangles and Exterior Angle Theorem
5. Applications

Overview

Lesson 7 Supported Standard

Common Core Focus Standards

There is only one supported standard throughout the topic B materials in this Module. That is:

G-CO.C.9
Prove theorems about lines and angles. Theorems include: vertical angles are congruent; when a transversal crosses parallel lines, alternate interior angles are congruent and corresponding angles are congruent; points on a perpendicular bisector of a line segment are exactly those equidistant from the segment's endpoints.

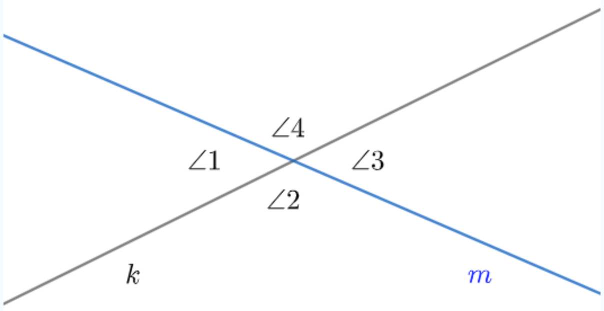
Student View: Slide 2

Lines Intersecting at a Point

Review of Facts

KNOWLEDGE POINTS: On two intersecting lines

Lines k and m intersect at point P (not shown). There are four angles with a common vertex at P. Angles 1 and 3 form a ***vertical pair,*** as do angles 2 and 4.



FACTS:

- $m\angle 1 = m\angle 3$
- $m\angle 2 = m\angle 4$

REASON: If two angles form a vertical pair, then they have equal measure.

ABBREVIATION: If $\angle A$ and $\angle B$ form a vertical pair, write "A and B are vert.$\angle$'s."

HOW TO STATE: (as a reason in a proof or exercise) If you need to state ***why*** $m\angle A$ and $m\angle B$ (vert.$\angle$'s) are equal, write "Vertical angles are equal in measure."

Instructor Notes

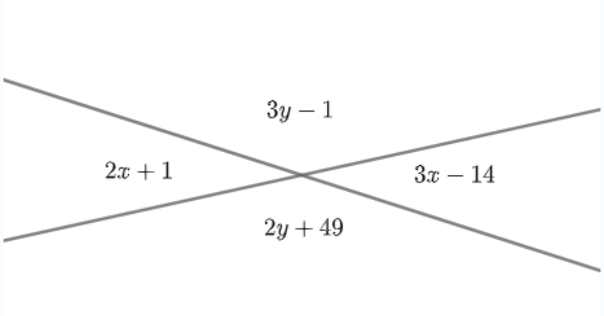
- We introduce the term "vertical pair" to help students understand that there is first a condition that must be met before they can simply say "vertical angles are equal in measure." Some students think that just any angles having a common vertex are "vertical".
- While there are only two vertical pairs, there are four "linear pairs" formed by two intersecting lines. They know about linear pairs so it is a good idea to ask them to find all the linear pairs here as well.
- They should be familiar with the abbreviations here, but make sure to review for completeness.
- For most of this course angle measure is always in degrees.

Student View: Slide 3

Lines Intersecting at a Point

Practice

In the diagram, both lines in the plane intersect. Each algebraic expression represents a degree measurement.†



1. Find the value of $x+y$. State your reasons.
2. What is the measure of either of the smaller angles?
3. What is the measure of either of the larger angles?
4. How could you have obtained the measure of a larger angle from the measure a smaller angle without finding y?

Instructor Notes

† The alternative would be to write, for example, $(3y-1)°$, which we will do later. The author was a bit too hasty in putting the diagram together. Nevertheless, this is a common way of specifying the meaning of algebraic expressions in diagrams and should not be a source of much confusion. Simply, draw attention to the difference between, for example, the meaning of $3y-1$ — representing a measurement of degrees, and the variable y — representing a generic real number.

1. $x+y=65$. The larger angles are vert.∠s, as are the smaller. Therefore their measures are equal. We therefore have $2x+1=3x-14$. Solving, $x=15$. Similarly, $3y-1=2y+49$ so that $y=50$.
2. Plug $x = 15$ into either x–expressions to get 31 for the smaller angles.
3. Plug $y = 50$ into either y–expressions to get 149.
4. A larger angle forms a linear pair with either of the smaller, so their angle sum is 180. Once we find that the smaller angle is 31, subtract from 180 to get the result.

- They have seen that linear pairs (angles on a line) are supplementary and the supplementary angles add to 180.
- Since all measures of all angles in this course (for the most part) are in degrees, we will suppress always writing the degree symbol except when necessary to avoid confusion. Alert the student to this convention and note that they do not need to always supply a degree symbol either. (For this course)

Student View: Slide 4

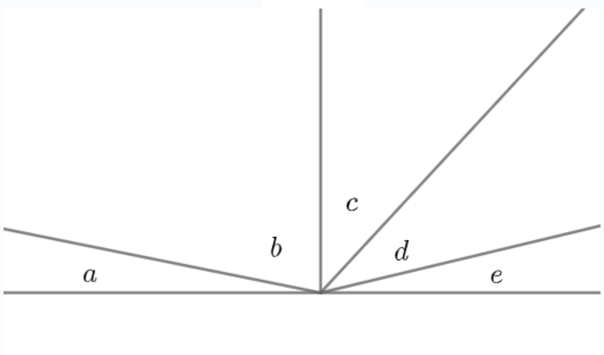
Several angles on a line

Review of Facts

KNOWLEDGE POINT: Consecutive adjacent[†] angles on a line
Suppose a sequence of n consecutive adjacent angles satisfies the two conditions.

- The interiors of the angles are all disjoint.
- The angle formed by the first $n - 1$ angles[1] and the last angle form a ***linear pair.***

Then, the sum of all the angle measures is 180.



The figure above shows the case where $n = 5$ measurements (in degrees) indicated by the variables.

$$a + b + c + d + e = 180$$

ABBREVIATION: $\angle$s on a line

HOW TO SAY[2]: Consecutive adjacent angles on a line sum to 180.

Instructor Notes

[†] This concept was defined earlier make sure to clarify that the student understands this.

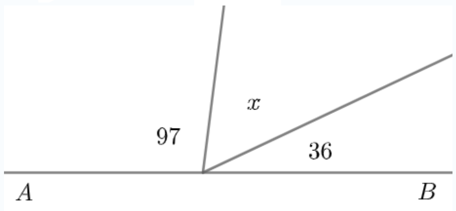
1. Well, this is not technically correct. Since an angle is a union of rays, we should say "the angle formed by *the union of* the first $n-1$ angles." But we will go with the abuse of language to avoid being even more cumbersome.
2. We are deliberately omitting all the extra verbiage of the last section. We hope the point is understood.

Student View: Slide 5

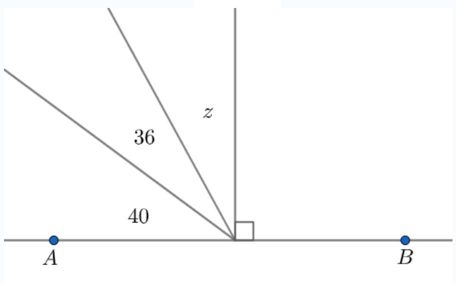
Several angles on a line

Practice†

EXERCISE 1. Find the value of x. All angles are on line AB. State your reasons.



EXERCISE 2. Find the value of z. All angles are on line AB. State your reasons.



Instructor Notes

† They are also familiar with the “angle addition” postulate, as well as fact linear pairs are supplementary. We use these facts in this practice implicitly.

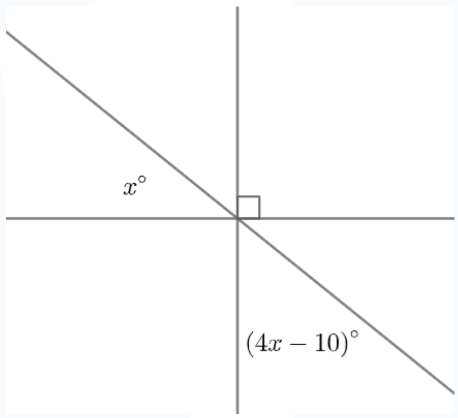
1. $x = 47$. Consecutive adjacent angles on a line sum to 180.
2. $z = 14$. Consecutive adjacent angles on a line sum to 180.

Student View: Slide 6

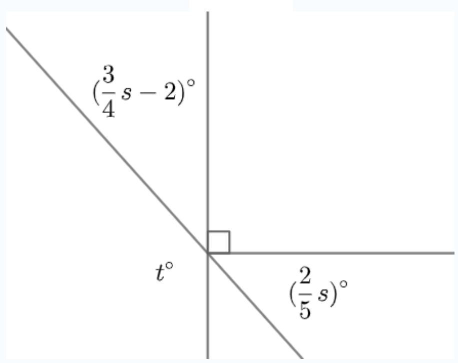
Several angles on a line

Practice

EXERCISE 3. Find the value of x. State your reasons.



EXERCISE 4. Find the value of s and t. State your reasons.



Instructor Notes

⇒ Note here the use of the degree symbol °. We are using that here to differentiate the algebraic expression that results in a degree measurement. So here the degree symbol serves to differentiate between the values of x which are real numbers, and the degree measures yielded by functions of x.

1. $x = 20$. Vertical angles are equal in measure. Consecutive adjacent angles on a line sum to 180.
2. $s = 80$ and $t = 122$. Consecutive adjacent angles on a line sum to 180.

Student View: Slide 7

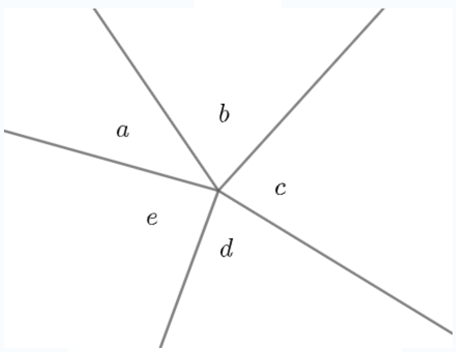
Angles at a point

Review of facts

KNOWLEDGE POINT: Suppose three or more rays all have the same vertex and their interiors do not overlap. Then sum of the measure of all angles formed is 360.

ABBREVIATION: $\angle$s at a point

HOW TO SAY: Angles at a point sum to 360.



In the figure there are five measurements* indicated.

$$a + b + c + d + e = 360.$$

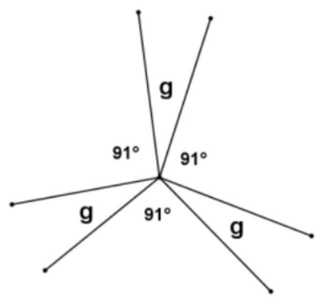
Instructor Notes

* We again abuse notation slightly. Sometimes when we mean angle a we will write simply a in the drawing. For this reason we said that "measurements" are represented here. This sort of context dependent abuse of notation is common in textbooks and students should get used to using "context" clues to determine how to interpret the labels. A similar problem happens in algebra when we have to make the distinction between $f(x)$ — the value of the function f at x, and the function f. Nevertheless we will try to be as consistent as we can.

Student View: Slide 8

Angles at a point

Practice



EXERCISE 1: In the figure above, g represents the ***number of degrees*** in each of the three congruent angles. Find the value of g. State your reasons.

Did you notice what happened? The style of the figures changed. The figure you are seeing in this slide is a screenshot from the New York material. I am experimenting to see how much less time it takes if we use this method. Up until this point, I have been either designing my own figures in Geogebra or LaTeX and changing some of the features while still retaining the main idea. (For the most part)

OBSERVATIONS:

1° This might violate copyright laws if the exact figures and problems are included in published work, so eventually similar problems will have to be crafted and coded.

2° There are many conceptual issues with the actual New York material as it actually stands. This figure is a great example!

3° Note that in their convention gis a ***label*** for an angle. In fact when the present the solution they even write $m\angle g = 29°$. There is a problem here. Which angle g?Those three angles are congruent but they are not ***the same*** angle.

4° In my convention, we would say that gis the angle measure in degrees and leave off the little ° symbol on the known angles.

5° As we move forward we need to take note of these details.

6° The time we save using screenshots might be offset by the time we have to spend modifying the problem to use the screenshot!

Instructor Notes

SOLUTION:

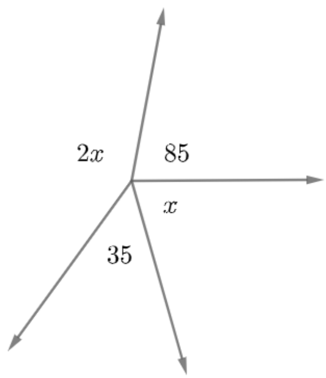
Angles at a point sum to 360.

g = 29.

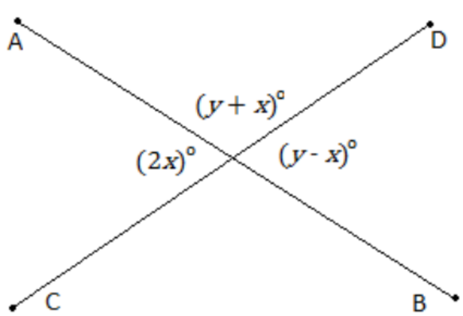
Student View: Slide 9

Angles at a point

Practice



EXERCISE 2: In the figure above, all rays have a common vertex. Each expression represents ***how many***† degrees are in each angle measurement. Find the value of x. State all of the reasons you use.



EXERCISE 3: The figure above shows intersecting segments $\overline{AB}$ and $\overline{CD}$. The angle measures†† are given here in terms of algebraic expressions. Find x and y. State the reasons for your answers.

Instructor Notes

† Again we are using this convention that we will use the language of the problem to avoid confusion. Be sure to point this out. So x is a number of degrees as is 35.

†† The difference here is this: say E is the intersection point. Then we would properly write $m\angle AED = (y + x)°$. But if we hold to convention that angle measure is ALWAYS in degrees, unless otherwise stated, we could just as well write $m\angle AED = y + x$. This is why the degree symbol is a bit redundant.

SOLUTIONS:

EXERCISE 2: $x = 80$. Angles at a point sum to 360.

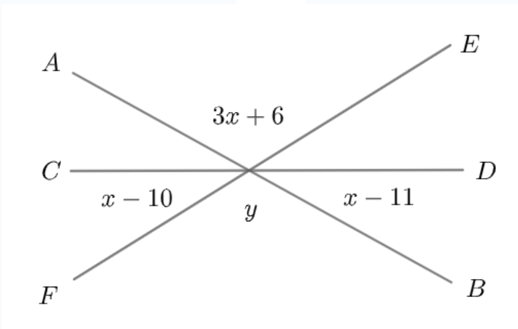
EXERCISE 3: $x = 30$ and $y = 90$. Vertical angles are equal in measure. Angles at a point sum to 360.

Student View: Slide 10

Angles at a point

Practice

EXERCISE 4: In the figure below, segments $\overline{AB}$, $\overline{CD}$, and $\overline{FE}$ are all concurrent†. Each of the algebraic expressions represents how many degrees are in each angle measurement. Find the values of x and y and state all the reasons you to find the answer.



Instructor Notes

† This is a great chance to review this topic from the last couple lessons. Three or more lines (or segments) are concurrent when they share the same point of intersection.

Why is do we need labeled segments?

SOLUTION: $x = 39$ and $y = 123$. Vertical angles are equal in measure. Angles at a point sum to 360.

Do we need to show the algebra?

Student View: Slide 11

Angles in a triangles

Review of facts

KNOWLEDGE POINT: The sum of the measures of the angles in any triangle is always 180.

EXPLANATION: We will chart out a series of steps you can go through on your own to verify this.

Instructor Notes

This is an extremely widely used result. It is important that students have some inclination of how it is related to work we have done so far. The "explanation" we provide here is not a rigorous proof, but merely a way of connecting this familiar idea to the most recent review exercises.

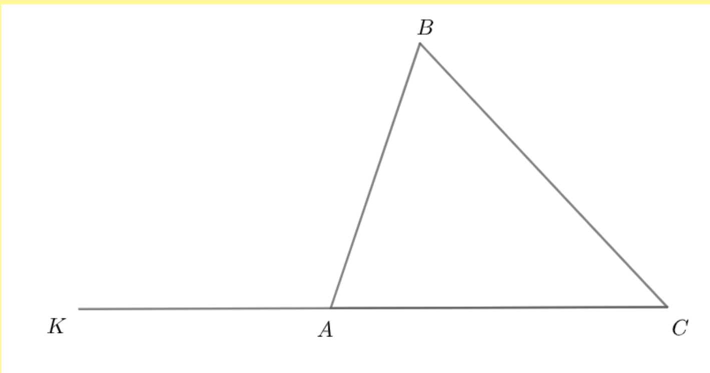
An animation would be helpful.

Lesson Design Specifications (Please Read)

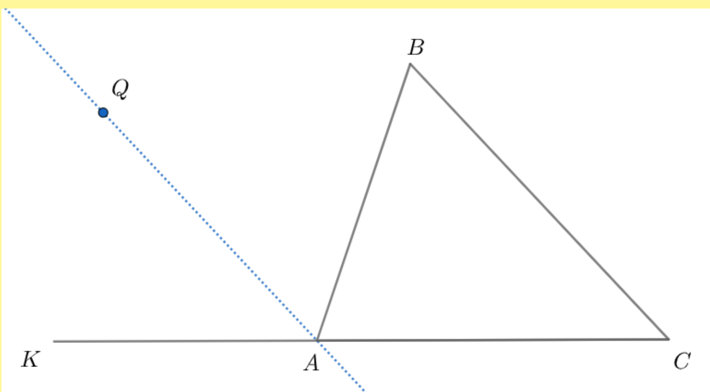
Guided Practice:

At this point student and teacher will do one of the following:

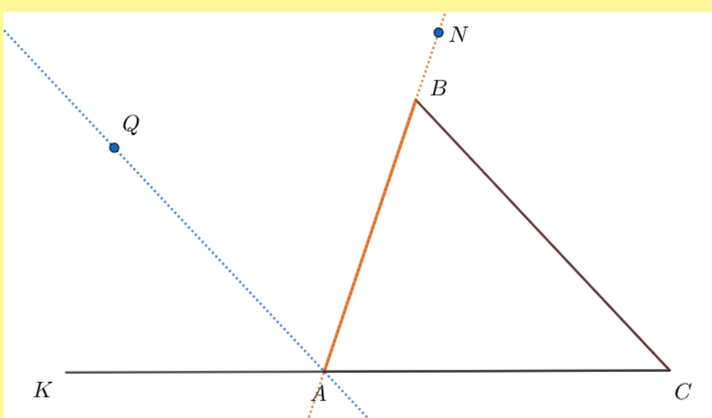
- Navigate to Geogebra on a shared screen **OR**
- Launch an app that opens up a shared window for interactive geometry **OR**
- Share a whiteboard with appropriate drawing tools.



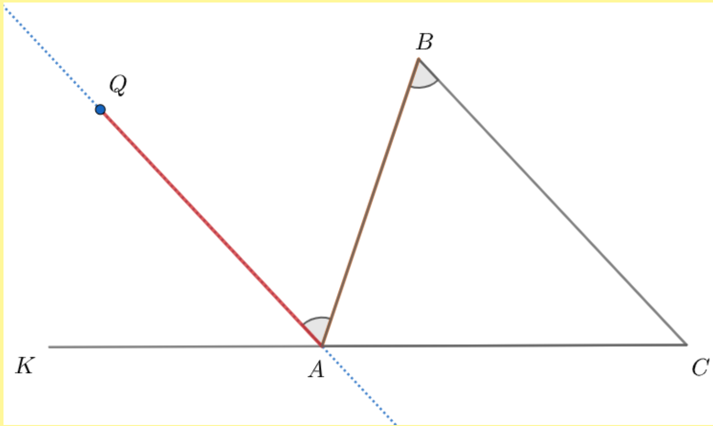
STEP1: Start with $\triangle ABC$ and extend segment AC as shown.



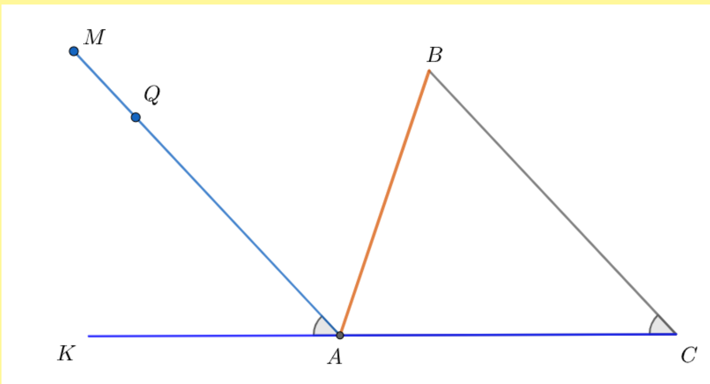
STEP 2: Draw line AQ parallel to $\overline{BC}$ as shown.



STEP 3: Locate N on the same side of line AQ as B so that $\angle QAN \cong \angle CBA$. Observe: B lies on line AN. (We can do this for example by verifying that $AN = AB + BN$.)



STEP 4: It follows that $\angle KAQ$, $\angle QAB$ and $\angle BAC$ are all angles on a line. (We needed to establish that B was in fact on line AN in order to make this statement.) Therefore the sum of their measures is 180. We will now show that $\angle KAQ \cong \angle ACB$.



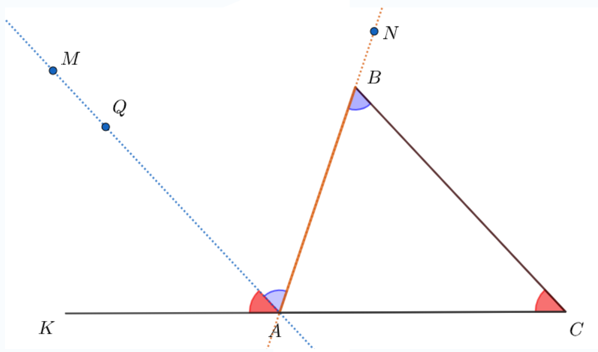
STEP 5: To verify that $\angle KAQ \cong \angle ACB$ we follow a similar procedure as in **STEP 3**. Locate M on the same side of $\overline{CK}$ as B so that $\angle MAK \cong \angle BCA$. Observe that Q lies on segment AM.

We now return to the main instructional sequence.

Student View: Slide 12

Angles in a triangle

Review of facts



SUMMARY: We have shown that $\angle MAK$, $\angle QAB$ and $\angle BAC$ are all† angles on a line and therefore the sum of their measures is 180. Since $\angle MAK \cong \angle BCA$, and $\angle QAB \cong \angle CBA$ we can conclude that the sum of the measures of angles $\angle BCA$, $\angle CBA$, and $\angle BAC$ is also 180! These are the angles in the triangle.

In our next lesson, we will show how to give a much simpler explanation without copying angles.

Instructor Notes

† Actually not quite. We showed $\angle QAK$, $\angle QAB$ and $\angle BAC$ were angles on a line. Since Q is on $\overleftrightarrow{AM}$ the statement we use here is valid. Some clever students might comment on this.

Student View: Slide 13

Angles in a triangle

Review of facts

VOCABULARY: Let $\mathcal{T}$ be a given triangle in the plane.

1° If at least two sides of $\mathcal{T}$ are congruent, $\mathcal{T}$ is said to be ***isosceles***.

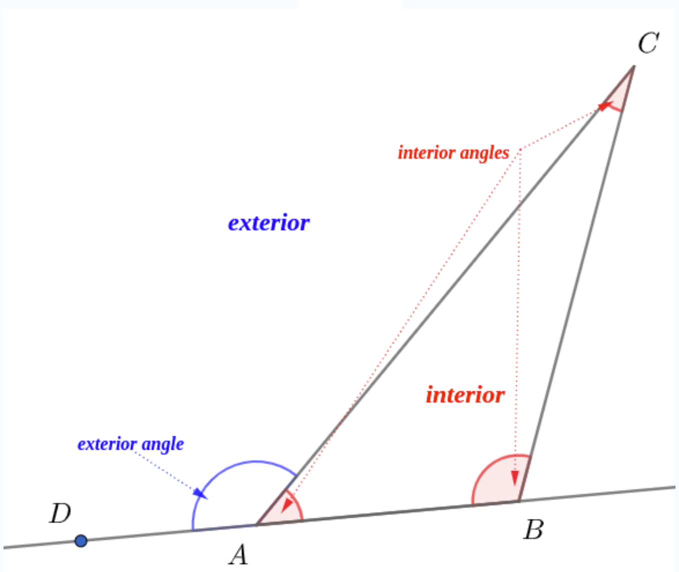
2° If all three sides of $\mathcal{T}$ are congruent, then $\mathcal{T}$ is said to be ***equilateral.***

3° The ***interior*** of a triangle is the intersection of the interiors of each of its angles which are called ***interior angles***.

4° A point is on the ***exterior***† of a triangle if it is any point not in the interior or on its sides.

5° $\angle DAC$ is called an ***exterior angle***†† if D is on line AB and $\angle BAC$ and $\angle DAC$ form a linear pair.

The figure below should help visualize 3°, 4° and 5°.



In the figure, $\angle B$ and $\angle C$ are said to be ***remote interior angles*** relative to the exterior angle $\angle DAC$. *

Instructor Notes

These vocabulary terms should be familiar. We just review them here for completeness.

† A triangle divides the plane into two disjoint sets: one bounded, the other unbounded. The bounded region is convex, while the unbounded region is not. An easy, but useful observation.

†† This forces the condition that A must be between D and B. A good observation to point out. What goes wrong if that condition is violated?

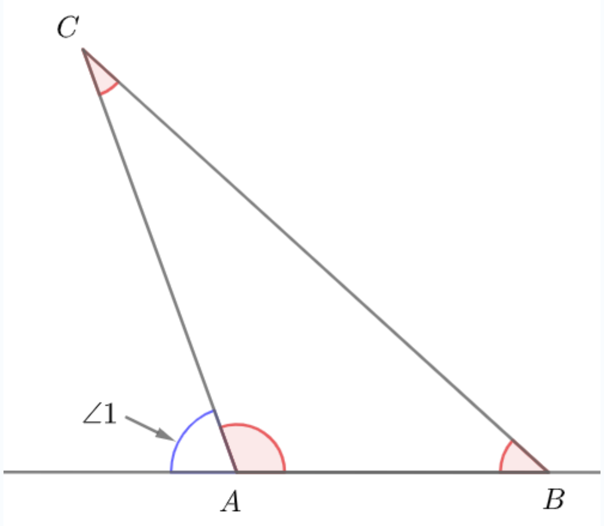
* The exterior angle $\angle KAB$ formed along side AC also shares $\angle B$ and $\angle C$ as remote interior angles, so it is important to specify which exterior angle we are speaking of. Their measures however are the same.

Student View: Slide 14

Angles in triangles

Review of facts

Knowledge Point: *Exterior Angle Theorem*
In any triangle the measure of an exterior angle is always the sum of its two remote interior angles.



Here, according to the figure, by the Exterior Angle Theorem we can conclude that

$$m\angle 1 = m\angle B + m\angle C.$$

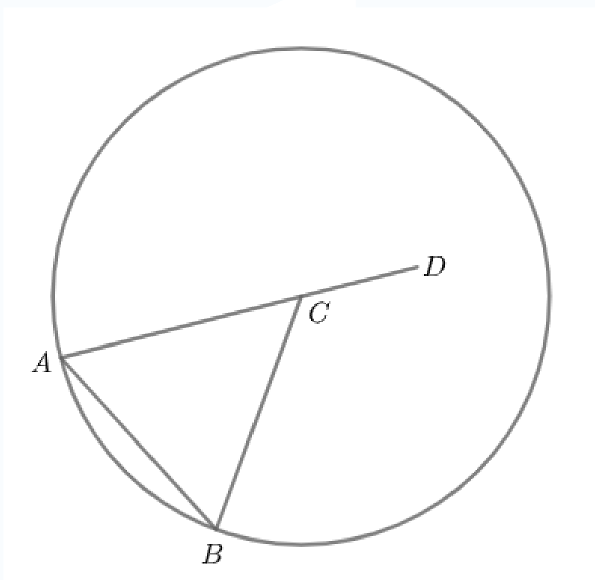
Instructor Notes

- It is fine to label angles like this when there is no ambiguity.
- $\angle 1$ and $\angle A$ form a linear pair, so there should be no confusion here.
- Later in the course, we will give a proof of this fact. We just accept it as a given at this point to work with it.

Student View: Slide 15

Applications

Relationship with circles



In the figure, circle C has center at C. A and B lie on circle C. $\overline{AD}$ is part of the diameter and AD is larger than the radius of circle C. If $m\angle DCA = 122$ find the measures of all the interior angles of $\triangle ABC$. State the reasons for each of your calculations. (Hint: the base angles of an isosceles triangle are congruent.)

Instructor Notes

1° $\angle DCB$ is exterior to $\triangle ABC$ (Definition)

2° $m\angle DCB = m\angle A + m\angle B$. (Exterior angle theorem.)

3° $m\angle A = m\angle B$. (Hint, and cong. $\angle$s have the same measure)

4° $2m\angle A = m\angle DCB = 122$. (Substitution and given)

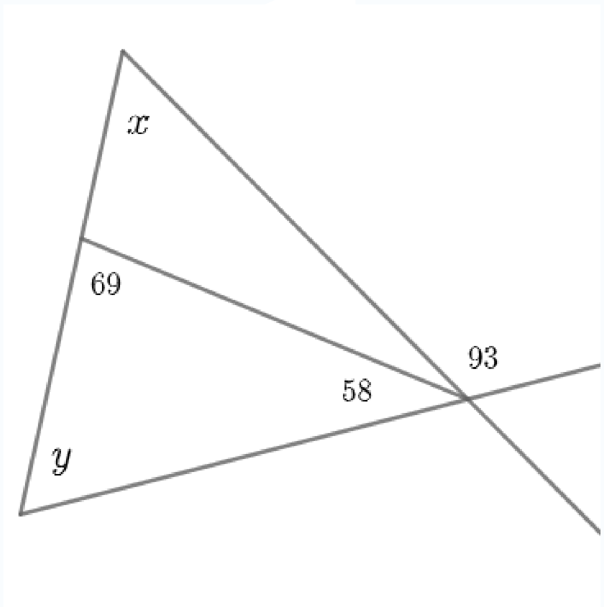
5° $m\angle A = 61 = m\angle B$ (Division)

6° $m\angle BCA = 58$ (Either sum of angles in a triangle or supplementary angles.)

Student View: Slide 16

Applications

Relationship with internal triangles



In the figure above, all segments which appear to intersect do intersect. All algebraic and numeric quantities represent how many degrees are in the angle measurement. Find x and y. State your reasons for your calculations.

Instructor Notes

1° $93 = x + y$ (Exterior angle theorem)

2° $y + 69 + 58 = 180$ (Sum of angles in a triangle)

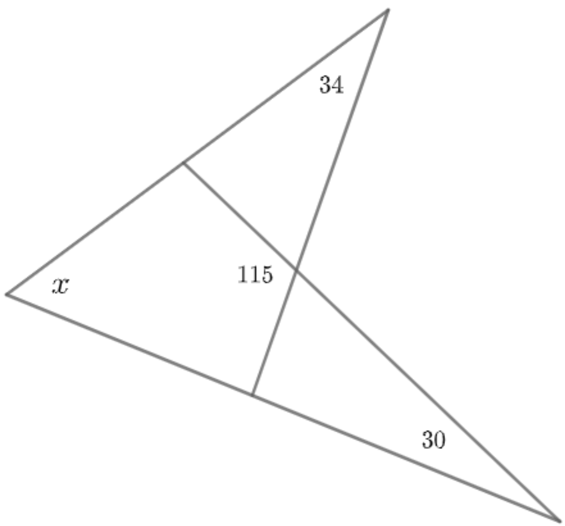
3° $y = 53, x = 40$ (Arithmetic and 1°)

Student View: Slide 17

Applications

Overlapping triangles

In the figure both large triangles "overlap". Segments which appear to intersect really do. The algebraic and numeric quantities represent the number of degrees in the angle measured. Find x. State your reasons.



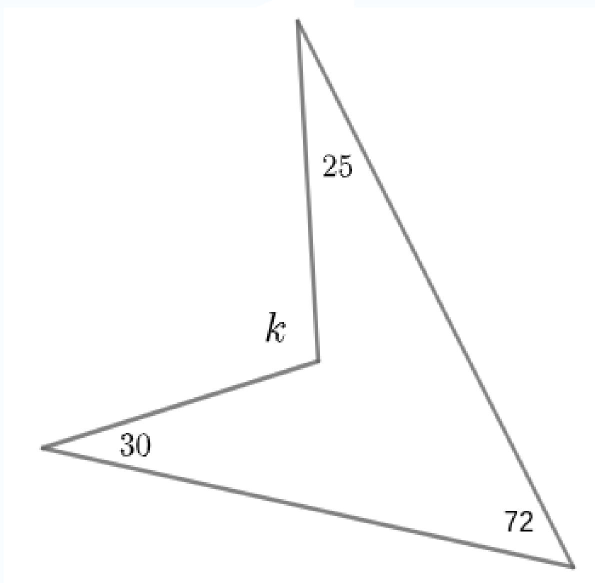
Instructor Notes

1. $x = 51$.
2. The measures of supplementary angles add to 180.
3. Sum of the measures of the angles in a triangle.

Student View: Slide 18

Applications

Angle chasing and "helping" lines



In the figure both large triangles "overlap". Segments which appear to intersect really do. The algebraic and numeric quantities represent the number of degrees in the angle measured. Find k. State your reasons.

Instructor Notes

1. $k = 127$.
2. Draw helping lines to make the figure look like the one in Slide 17.
3. Use the same reasoning in reverse to find k.